Technical Note — Government Override × Contracted Demand Runner (v2)

1) Background & Intent

This runner applies two post-model overlays to an already-scored MPI cohort: Government Override (α) and Contracted Demand (β), with an interaction knob (Θ). It does not re-fit Bayes or Cox engines. Instead, it simulates counterfactual shifts in project probabilities and expected values.

2) Inputs

- Required: p_bayes, p_cox, blended_prob, EV_disc (preferred) or EV_cum
- Optional: merchant_share, contract_coverage_share, ppa_flag, gov_ofstake_flag
- Pass-through: company, project

3) Method

Weight estimation (no refit):

w_bayes = clip((blended_prob - p_cox)/(p_bayes - p_cox), 0,1)

Override lever (α):

Anchors at α=1 use OR≈2.12 and HR≈1.65. Bayes logit shifted, Cox survival powered, then re-blended. Probability capped at 0.90.

Contracted Demand lever (β):

Coverage scales OR and HR anchors (defaults k=0.75, h=0.50). Applied per project, then re-blended.

Interaction (Θ):

Sequential composition in probability space, with optional logit bump scaled by theta_scale. $\Theta<0$ = substitution; $\Theta>0$ = complementarity.

Expected Value (EV):

EV_final = (EV_base/p_base) × p_final. Uses EV_disc if present.

4) Outputs

- override_contract_overlay_full.csv — per-project overlays
- override_summary_by_alpha.csv — α-only summaries
- override_contract_grid_summary.csv — $\alpha\times\beta$ grid with Θ
- override_anchor_sensitivity.csv — sensitivity by anchor bands
- override_run_audit.json — reproducibility metadata

5) Replication Checklist

- Provide input CSV with required columns
- Declare scenario knobs: alphas, betas, theta, cap_p
- Run via Colab single-cell or CLI
- Archive inputs and all outputs
- Log CONFIG and code hash (audit JSON written automatically)

6) Interpretation Tips

Use the four-cell lens: (Status vs Override) × (β=0 vs β>0). Crossers ($\geq$0.50) often carry the investability story. Complementarity vs substitution can be assessed with Θ parameter sweeps.

7) Limitations

Counterfactual only; stylized scenario dials; not forecasts.